Understanding the tomato-cutting project

A detailed, plain-language history

From the original research ambition to cutting-model development, robot integration and vision-guided knife pickup.

Record reviewed through 14 November 2026.
Prepared for the project researcher.
Robotics basics assumed; simulation and learning explained step by step.

Read this as an explanation of the evidence, not a claim that the entire system is finished. The latest pickup work was still in progress at the review cutoff.

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Understanding the tomato-cutting project

A step-by-step history of the work, the failures, and the decisions

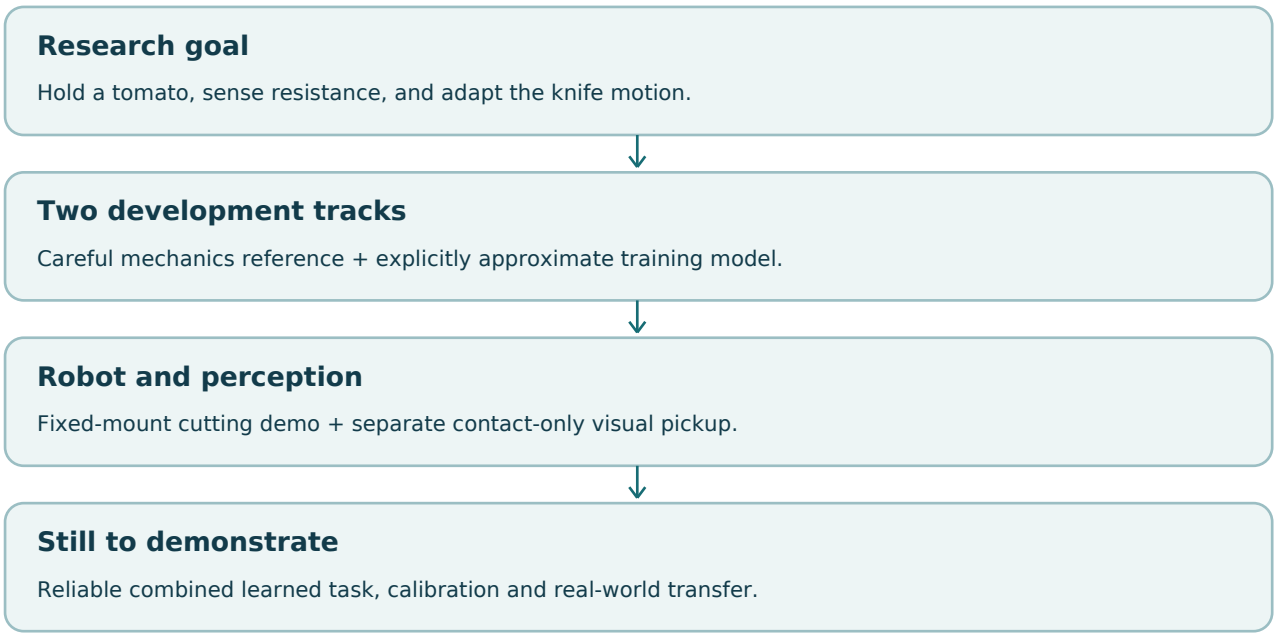
This document is for you: a readable explanation of how the project developed, rather than a paper claiming that everything is finished. It covers the accessible research record from the early October 2026 planning documents through the November 14 repository update and latest pickup-chat messages inspected for this report.

The story has several branches. Robot setup, knife grasping, cutting physics, numerical accuracy, performance, visual appearance and learning developed at different speeds. A successful result in one branch did not automatically complete the others. Keeping those branches separate is the key to understanding the project.

The account is reconstructed from saved plans, result reports, experiment folders, the repository and the continuation chat. Some early steps have only summary records. Dates on older living documents sometimes reflect when the document began, not every later edit. Where the sequence is known but the exact date is not, this document says so. It does not pretend that every command or abandoned thought can be recovered.

Source labels in square brackets lead to the evidence appendix. That appendix also contains an index of individual experiment reports and plain-language explanations of the later version sequence. The underlying reports remain the authority for exact settings and acceptance limits.

1. The project in one view



1.1 The original ambition

The original ambition was a two-arm robot that could hold a tomato and slice it intelligently. One arm would stabilize the food. The other would move the knife. The controller would use camera information and contact or force information to adjust to uncertainty: a tomato may be soft, firm, slippery, differently shaped, or positioned unexpectedly. [S01]

The early research question was deliberately ambitious. It combined hierarchical residual reinforcement learning, multiple sensor types, compliant control and transfer from simulation to a real robot. These were proposed research objectives. They were not capabilities already demonstrated by the robot.

1.2 What we have now

The project now has a large body of source code, diagnostics, simulation scenes, experiment reports and preserved failure evidence. It has an approximate tomato model that completed restricted simulated cutting tasks, a fixed-mount robot/kitchen cutting demonstration, and a separate camera-guided contact-only knife pickup prototype. Those are important but different achievements. [S12-S16]

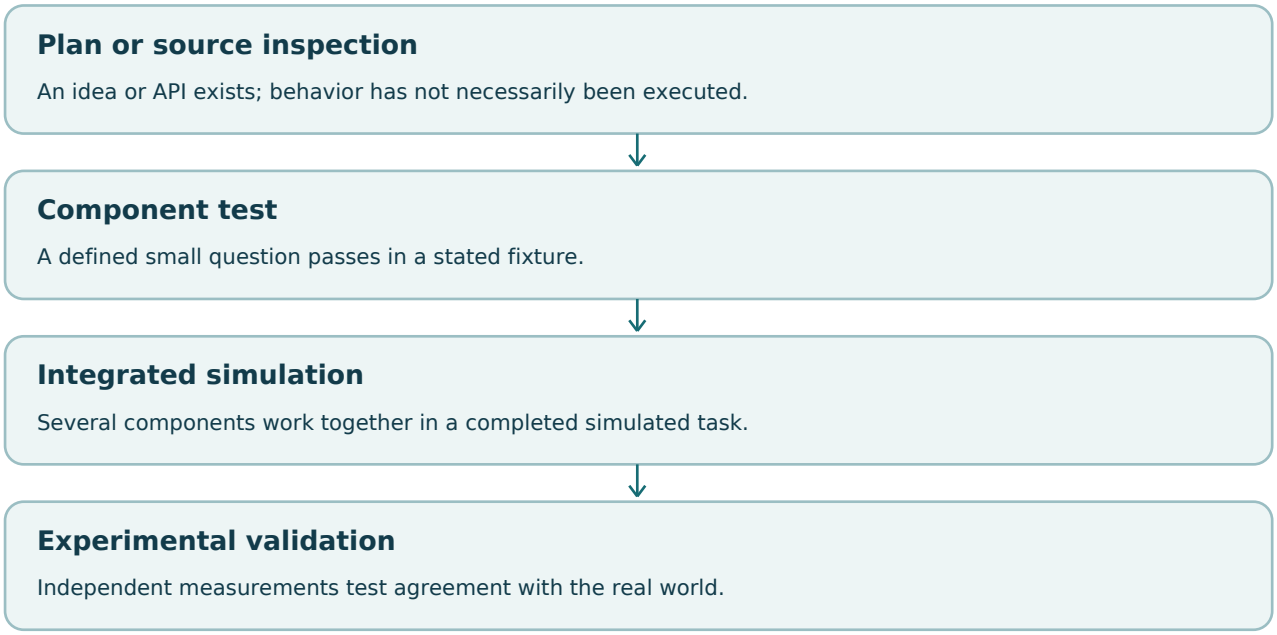
The stricter cutting-physics branch remains an unfinished numerical research effort. The approximate model deliberately accepts material and discretization limitations. The pickup branch has a strong first result on a limited knife setup, but broader shapes and resting poses remain under development. A complete learned, calibrated, dual-arm real-world tomato-cutting system is not established.

1.3 Why the work looked like repeated detours

The project repeatedly exposed a hidden prerequisite. Before learning cutting, the material had to separate correctly. Before trusting separation, contact and energy had to be checked. Before running many learning episodes, the simulator had to be fast enough. Before asking a hand to carry the knife, the grip had to work physically. These were dependencies, not interchangeable checklist items.

However, the history also shows a real scope problem: trying to make a very accurate custom cutting simulator and a sophisticated robot-learning system at the same time created many possible research projects. Later decisions increasingly separated an accurate reference track from an approximate training track.

2. How to read the evidence



2.1 A plan is not a result

A plan describes intended work. Implemented code means an attempt exists. A unit test checks a small defined behavior. A completed simulation demonstrates behavior in that simulation. A calibrated physical experiment tests agreement with the real world. These are different levels of evidence.

For example, a test can prove that an energy counter updates correctly without proving that the knife actually produces a realistic cut. A scene can show a knife in a hand without proving that finger friction supports it. The report will keep making these distinctions because they explain many apparent contradictions in earlier updates.

2.2 A coupon is a small test specimen

The project often tested a small block, interface or pair of pieces instead of a whole tomato. This is called a coupon. It isolates one question: whether contact pushes in the right direction, whether a damaged interface stays damaged, or whether one fragment can move without dragging another.

Passing a coupon is useful. It is easier to diagnose than a full robot scene. But the whole system adds shape, support, contact combinations, longer time histories and more computation. A coupon pass is a building block, not a full-system certificate.

2.3 Three meanings of a successful cut

A geometric cut means the representation has been divided along a selected region. A mechanically independent cut means the resulting pieces no longer transmit unintended internal forces. A work-paid physical-model cut also requires a declared fracture rule to spend irreversible work before connections disappear.

Even that last category is still a simulation result. Realistic tomato cutting additionally needs measured material behavior, appropriate blade contact, and experimental validation. The approximate model later achieved work-paid simulated cuts without meeting the stricter continuum-accuracy requirements.

2.4 Why failed runs remain in the record

A failed trial tells us which assumption did not work. Deleting it would make a later improvement look more reliable than it was. The project therefore retained failed configurations, logs, source snapshots and rejected candidate states.

This also protects comparisons. Once a test case has been used to tune a controller, it becomes a development or regression case. It should not later be described as an untouched independent test.

3. Early October: research planning and robot foundations

3.1 The first research architecture

The October 5 decision log placed planning, literature, experiments, simulation, learning, hardware and manuscript work in one Git-based research workspace. Structured research-state files were intended to hold questions, decisions, experiment definitions and claims. [S01]

This was an organizational foundation. It aimed to make the work traceable across long chats and multiple tools. It did not mean that all proposed components had been implemented.

3.2 The proposed arm roles

The early embodiment described ARX AR5-L6 arms and LinkerHand O6 dexterous hands. The left side would hold the tomato and the right side would slice. Later practical work concentrated heavily on the right arm, knife and cutting environment.

The early documents also proposed high-rate tactile or force feedback. Hardware encoder interfaces, sensor variants and calibrated limits were not all verified in the later audit. Therefore, those proposed sensors must not be confused with channels actually available to a trained controller.

3.3 Why residual learning was proposed

Residual learning means starting with an understandable baseline movement and learning a correction. For example, a baseline may move the knife downward while sawing; a learner adjusts speed or compliance when resistance changes.

The attraction was a smaller search problem than asking a network to discover every joint movement from nothing. The early proposed action included Cartesian velocity adjustments and compliance gains. This was a design direction, not evidence that residual cutting had been trained successfully.

3.4 Early platform assumptions needed later verification

The first planning documents selected Isaac Sim, Isaac Lab and PhysX deformables and anticipated efficient GPU training. Some statements in those documents were expectations about cutting and parallelism, not demonstrated installed capabilities.

Later source inspections and numerical experiments showed why those expectations needed testing. A simulator supporting deformable objects does not automatically support the particular fracture law, contact fidelity, persistent cuts and robot coupling required here.

3.5 Joint stability and virtual inertia

An October 7 decision approved added joint armature and default proportional-derivative gains for simulation stability. Armature adds virtual rotational inertia to a joint; it can make numerical motion less explosive when light fingers connect to heavier links. [S01]

This is a modeling choice with a cost. It changes dynamic response, particularly at high frequency. The record explicitly noted a simulation-to-real risk and proposed actuator randomization as a mitigation. Approval of a stable simulated joint is not proof that its real dynamics match.

4. October 19-era audit: could the laptop support the workflow?

4.1 Counting and checking the actual model

The phase audit identified seven independent right-arm joints, six independent hand joints and five passive hand followers. It also checked software and a CUDA smoke test. Tactile channels were excluded where their real interfaces were unverified. [S02]

A follower joint is not another independent command. This distinction matters when defining actions, observations and control limits. Counting visible joints alone can produce the wrong controller interface.

4.2 The resource benchmark

A minimal scene combined the robot, board, countertop, free knife, an RGB-D camera and a small learning workload. RGB-D means a color image plus a depth image. The long run lasted about 30 minutes, completed 277 resets and performed 12 tiny PPO updates. [S02]

PPO is a reinforcement-learning optimization algorithm. In this test, it used a posture reward and an untrained image encoder. The purpose was to exercise software, memory, reset and checkpoint behavior. It did not learn knife recognition, pickup or cutting.

4.3 What the measured numbers meant

The long probe recorded roughly 7.747 GB peak process RAM, 2,658 MiB peak total GPU use and 9.83 actions per wall-clock second. Post-warmup memory growth was small over that measured window. These were encouraging feasibility measurements for one environment, not evidence of multi-day stability or a large parallel training capacity.

A checkpoint is a saved state used to continue work. The benchmark checked restoration of optimizer and random-generator state and a deliberate stop/resume path. It restarted a fresh physical episode; it did not demonstrate exact restoration of every hidden in-flight contact state.

4.4 Safeguards became part of the workflow

Startup reserves, checkpoint reserves, hard memory stops and a shared controller lock were added or verified. The lock prevents two controllers from issuing commands into the same shared setup. These protections were there to preserve the laptop and evidence, not to make a failed physical test pass.

The decision was to continue a bounded one-environment prototype while keeping full task training gated. Native integration checks later passed in their scope, but physical pickup remained unresolved.

5. The first knife-grasping campaign

5.1 Why touching the handle was not enough

The early central-pickup tests used a free knife lying on the board. There was no attachment or in-episode knife teleport. The arm approached through joint drives and physical contact. Initial opposing pinch contact was achieved, but the knife did not remain securely carried during clearance. [S03]

The long knife also constrained placement: the recorded knife length was approximately 43.8 cm on a board approximately 46.9 by 29.9 cm. A grasp that looks reachable may still place the blade outside the safe board footprint.

5.2 Tilting created access, not an unsupported pickup

The team tried lifting the handle while the blade remained supported on the board. Early tests obtained about 15 mm handle rise, followed by a supported plateau. Later palm-seating work raised the handle to about 36.8 mm.

This was real contact-mediated motion, but the board was still carrying part of the load. It did not meet the unsupported hold requirement. The distinction prevented a visually impressive tilt from being reported as a completed grasp.

5.3 Seating the handle in the palm

The next idea was to preserve the thumb-index pinch while moving the wrist and other fingers toward a more secure wrap. Offline paths constrained selected contact points and normals, while the runtime controller checked tracking and drift.

A small final adjustment preserved the pinch with about 0.807 mm additional lateral motion. It did not establish palm contact or a three-finger wrap, so the system refused to lift. In other trials, contact vanished during motion before a large force spike. The cause was not conclusively reduced to one friction or collision issue.

5.4 Transferring load between fingers

Six bounded contact-transfer trials explored acquisition angles, ring clearance, middle-finger staging and rolling. Stable transfer to the additional fingers was not demonstrated. Other launches were refused by the memory reserve instead of proceeding under unsafe resource conditions.

The important geometric lesson was that curling fingers more does not always bring their useful inner surfaces closer to the handle. Depending on wrist and handle orientation, more flexion can move the contact surface away or push the knife against the board.

5.5 Offline grip checks changed the diagnosis

Offline geometry and static load checks found that some proposed layouts lacked useful opposing inner contacts or could not balance the knife's turning moment. A rolled, approximately edge-on terminal proposal had conditional static support under the assumed loads. [S03]

Static support is like balancing a wrench on paper: it tests whether available contact forces could resist weight and torque. It does not prove that the robot can reach that grip, establish it dynamically, lift without collision, and keep it during cutting. This distinction remains relevant to the November pickup work.

6. The first custom tomato model

6.1 Why a separate mechanics model was explored

The capability audit looked at installed source rather than assuming an engine could perform the required cut. Newton offered particle, spring and contact building blocks, but arbitrary fracture and supported coupling were not automatic. Coincident spring endpoints were also unsafe in a source path that divided by separation. [S04]

The initial design therefore became a separate, inspectable mechanics model. It used two prepared sides of a potential end-cap cut, linked by damageable interface elements. That made one cut easier to define, but it restricted the cut geometry in advance.

6.2 Nodes, springs and interface patches

Nodes are points carrying the model's motion. Springs resist changes between connected points. The interface was grouped into patches, each with a saved area and shared damage history. The original registry had 47 patches and 403 interface bonds. [S04]

The physical intent was that intact material transmits force, damage reduces its resistance, and fully broken connections no longer provide cohesion. A patch reaching complete failure is not the same as the entire tomato becoming disconnected: other patches may still bridge the two sides.

6.3 The first fracture-energy failure

The first scalar-to-bundle damage mapping did not spend the intended work. Opening energy was about 13.60% too high and shear energy about 64.20% too low. A scalar law that works for one ideal spring did not automatically work for a bundle of oblique finite-length springs.

The correction used the actual stored patch energy when evolving damage. Focused tests passed, and all 47 real interface patches subsequently passed opening and positive/negative shear checks: 235 cases under the unchanged 1% criterion. This qualified that law in those fixtures, not real tomato tissue.

6.4 Closed cut surfaces and sensible mass

Separate closed surfaces were created for the slice and remainder, with matched but independent cut-face nodes. Matching positions are useful for an initially closed interface; separate identities allow the faces to open later.

A provisional fixed surface-mass allocation was replaced by volume-based nodal masses. Directional spring stiffnesses were refined and small-strain compression/shear tests passed. These changes improved the model's internal consistency while leaving it explicitly synthetic and uncalibrated.

6.5 The large-deformation failure

Small-strain tests were not enough. Larger compression and shear inverted some local volume cells, even when the equilibrium force residual and some stiffness checks looked acceptable. An inverted cell means the local material mapping has folded through itself.

The repair added local volume energy to resist collapse. The model was now a hybrid spring-and-volume model, not a pure spring network. It passed 108 protected load stages across 18 compression/shear paths in the recorded study. This is why merely watching the outside of the tomato was insufficient.

6.6 Mesh refinement was its own investigation

Changing node spacing changed the response. A coarse mesh failed a selected comparison, and the first 4 mm geometry had duplicate-node and local-inversion issues. Reusing same-piece nodes and refining the remainder repaired those defects.

The slice still needed more internal sampling through its thickness. After that change, the refined slice family met the recorded 10% comparison target across the tested directions. These were limited mesh-sensitivity results, not universal convergence or calibration.

7. Contact, damage and the slow insertion campaign

7.1 Separating cohesion from compression

Cohesion holds two cut faces together while they are intact or partially damaged. Contact prevents them passing through one another when they close. A broken face should lose cohesion but should still collide with the other face.

An interface correction moved compression responsibility into unilateral contact. Unilateral means it pushes when surfaces try to overlap but does not attract separated surfaces. This avoided using tensile cohesion as an artificial compressive glue.

7.2 A first partial insertion

A surrogate wedge was coupled to irreversible damage in a shallow offline test. At 4 mm insertion, two patches had begun to damage but none had failed completely. The model then passed scoped unloading and supported-reclosure checks without healing the saved damage. [S04]

These solves were quasistatic: each step sought a force-balanced state rather than simulating all fast dynamics. They were useful for studying contact and work, but they were not a native robot cutting episode.

7.3 The between-vertex penetration defect

Checking only material vertices missed a blade crossing through the interior of a surface triangle. The vertices could all appear clear while the blade intersected the face between them. A finite-triangle counterexample stopped the deeper full-cut attempt.

The contact backend was corrected to reason about finite triangles and weighted forces. This is a general numerical lesson: sampling a few points does not certify an entire surface or volume.

7.4 Why a converged endpoint was not enough

The system also checked the swept path between accepted endpoints. A blade and material could be nonpenetrating at both ends while intersecting in the middle. Tight swept-clearance requirements forced smaller insertion increments near difficult contact changes.

Smoother cubic candidate paths reduced some measured penetration but did not rescue every interval. Rejected endpoints and trajectories stayed rejected. A nicer curve was not treated as permission to loosen physical acceptance limits.

7.5 Fixing the numerical search

The nonlinear solver had to balance forces while contacts activated and released. One search failure involved contacts predicting attractive loads that should have been released. Another involved a fallback that discarded useful off-diagonal stiffness information.

Separate search corrections preserved the physical energy and tangent while improving how candidate states were found. Independent audits still decided acceptance. This distinction matters: a numerical search aid is not a change to the material law, and an optimizer's approximate curvature is not a physical stability certificate.

7.6 What 8.84375 mm meant

One accepted branch reached 8.84375 mm insertion with two fully failed patches and one connected material component. Its withdrawal and supported reclosure were checked from that actual history. Damage was preserved, and the blade returned to zero insertion without pretending the remaining connections had disappeared. [S04]

The withdrawal branch and later forward-insertion branch must not be spliced into a fictional single successful cut. Each result belongs to its own saved predecessor chain.

7.7 What 9.638916015625 mm meant

The preserved October 27 forward checkpoint reached 9.638916015625 mm. Five of the 47 patches were fully failed; 42 still provided active connection, and the graph remained one component. It was progress in a difficult numerical continuation, not complete cutting or release. [S05]

An old dashboard sometimes displayed approximately 8.844 mm. That was a historical branch view, not the latest authoritative forward state. This is one reason later summaries needed source-bound evidence rather than reading a single visual progress label.

8. The bounded offline preparation period

8.1 Why the priority changed

The October 27 approved 15-hour preparation plan superseded further tiny-depth continuation. It redirected effort toward inventory, backend limits, units, history, resets, work accounting, guarded workers, mechanics evaluation and learning-interface preparation. [S05]

The aim was to make the next runtime stage concrete and testable, rather than spending the entire window advancing the knife by tiny fractions of a millimetre.

8.2 Contracts that prevented subtle mistakes

SI contracts fixed metres, seconds, newtons and joules. Frame contracts specified where forces and moments were expressed. Time contracts distinguished instantaneous force from accumulated impulse. History contracts prevented rejected trial steps from permanently damaging material.

Reset tests checked that episode history was cleared. Observation masks separated quantities a future robot could measure from privileged simulator information. Work ledgers separated stored energy, dissipated fracture work and external input.

8.3 The native-worker route was initially disabled

Separate worker/configuration code was prepared for small contact and two-piece tests, with execution guarded. Source inspection identified unresolved native integration and coupling issues. Static checks and process fixtures were not called native physics validation.

The preparation window ended on October 28. Quota-reading failures and an unfinished final handoff were preserved. The old automation was later deleted. Subsequent November runtime work belonged to later authorizations; it was not a silent restart of the expired overnight window.

9. October 31 to November 2: from isolated mechanics to native coupling

9.1 Preserving the work in Git

The project workspace was not itself the publishing Git checkout. A separate checkout was prepared to archive code, plans and evidence into TianjinUniversity. Large assets were left local and listed in a manifest. This protected the original working directories and other contributors' edits. [S18]

Authentication was initially successful as an account, but push access failed with HTTP 403. A token authenticates an identity; it does not grant repository rights that identity lacks. Later write access became available, and the snapshots reached the research branch.

9.2 Native contact and force-transfer tests

The next stage built increasingly demanding tests of geometry, force signs, reference frames, contact buffers and reaction transfer. A force applied at the wrong point can have the right magnitude but the wrong torque. An impulse divided by the wrong duration becomes the wrong force.

Small rigid probes, articulated examples and replay tests were used before claiming a mounted-robot result. Passing these tests established pieces of the coupling interface, not a complete cut.

9.3 Integrating the actual saved material

The November 1 report records the actual 4,032-node model with 33,996 bulk springs, 16,291 volume cells, 5,504 boundary triangles and 47 shared-history patches in the native implementation. It compared assembled forces and numerical derivatives against independent offline calculations. [S06]

Prescribed opening fixtures drove all patches while recording energy and support impulses. These were controlled material tests. The knife did not independently discover and cause every fracture event in those fixtures.

9.4 Keeping the original blade geometry

The saved blade had 74 vertices and 144 triangles. A constrained decomposition produced solid cells retaining that exterior, rather than replacing it with a larger convex hull. Surface triangles could then be clipped against blade volume to find intersections that vertex-only checks miss.

An earlier unconstrained decomposition failed and was rejected. The distinction matters because a simplified collision hull can change the effective sharpness and contact location even when the rendered knife looks unchanged.

9.5 The fixed mount was a deliberate simplification

The robot scene used an ideal fixed knife attachment in this branch. That allowed the team to investigate arm motion, material coupling and cutting without requiring a contact-only grasp to succeed first. The mount was disclosed; it was not evidence that the fingers could carry the knife by friction. [S07]

Later changes to stance, elbow placement, approach and withdrawal improved the visible route. A dry route can prove reach and clearance in that route, but it cannot prove loaded cutting or a stable free grasp.

9.6 The November 2 review corrected overbroad milestones

The review kept scoped component passes while reopening whole-system questions: sustained support, mixed material loading, mounted feedback, loaded reset, motor work and complete separation. It also kept experimental tomato validation pending. [S07]

A published force curve was considered a provisional reference. A plot alone cannot identify all material/contact parameters, and fitting points on one curve is not independent validation. The saved data-request draft was not sent.

10. November 3 to 7: visible robot motion and research cutting

10.1 Motion and grasp appearance developed alongside mechanics

The saved work includes kitchen stance, pickup demonstrations, overhand and wrapped-grip variants, and backward-elbow positioning. These changes made the sequence easier to view and execute in the scene. Their existence does not retroactively turn a fixed mount into a qualified finger-contact grasp.

This branch was useful because it separated scene and kinematic issues from material issues. A visible, reachable trajectory was a prerequisite for testing the cutting model under the actual arm.

10.2 Live fracture needed transactional history

The research slicing extension added a transaction mechanism: trial damage was computed separately, then committed only if the mechanical step passed. If the solve failed, the old state, work and damage were restored. [S08]

Prescribed patch opening agreed closely with the expected damage law, but loaded solves still failed. A decreasing energy value alone did not guarantee an acceptably balanced nonlinear solution. More iterations did not automatically repair the problem.

10.3 Smaller steps helped one control, not the entire cut

A very small timestep repaired a particular loaded control. A separate long downward route was kinematically reachable. Those were different tests: one checked mechanics and the other checked reach.

Neither established complete cutting. Collision, support, contact and force/work qualification still had to hold together during the actual route. The later mounted continuation remained slow and incomplete in the stricter research model.

11. November 8 to 10: DiSECt and the search for a usable backend

11.1 Profiling showed where time went

A short current-solver profile took about 62 seconds to simulate roughly 0.00667 seconds. Nonlinear solving and contact geometry dominated. Many device-to-host array calls also synchronized the GPU. File recording was a small measured cost. [S09]

This finding mattered: removing the kitchen display or deleting evidence logs would not solve the main bottleneck. Slow execution was also not proof of physical accuracy.

11.2 Stock DiSECt looked faster, but was not an adoption pass

The official prism example advanced five simulated seconds in roughly ten to eleven wall seconds in the local comparison. It was a different specimen, mesh, contact model and motion. Native cutting failure flags and shutdown crashes were also recorded.

Therefore the result was promising feasibility evidence, not an equal-accuracy speedup or a ready tomato backend. The later modified candidate must not be described as though it were the untouched stock implementation.

11.3 Transferring the research interface exposed a coverage gap

When the original 47-patch law was mapped to new cut geometry, a negative control found that 303 of 604 interface vertices had no nonzero participation in the mapped quadrature. Moving an uncovered vertex could change neither cohesive force nor energy. [S10]

The approved repair used full-surface sampling while retaining the total fracture budget and original patch laws. All interface vertices became covered. The distribution changed, and that change was documented rather than disguised as a pure speed optimization.

11.4 Coupled settling and contact remained expensive

A 100 ms assembled run had finite material state but was not settled. Larger-step and implicit candidates encountered convergence or physical-stop failures. Some algebraic optimizations matched a single step yet failed to make the complete system practical.

Later Section 4 work reached a compression stop before fracture. A separate softer material candidate used a literature-based prior, not measured calibration. Its coarse hold passed its own guards; the saved November 10 status still showed the refined comparison pending. This history does not support calling the strict DiSECt-derived tomato completed.

11.5 CRESSim was tested, not simply chosen

The public CRESSim-MPM release was built in isolation and ran fast on small fixtures. The team then tested what happened after the blade left. Nearby intended pieces shared motion through the background grid. With a blade held in a gap they could appear independent; with the blade absent, unwanted transfer returned. [S11]

In plain language, the cut could depend on the knife still acting as a barrier. That is not persistent separation. Stock CRESSim was therefore not adopted unchanged. The test also did not provide calibrated tomato cutting forces or anatomical tissue behavior.

12. November 10 to 13: the custom numerical research sequence

12.1 Why so many versions existed

The v01-v27 series tested different numerical representations and repairs. It was not 27 successful tomato simulators. Some versions passed a small component and exposed the next failure; others were rejected immediately. Appendix B explains each recorded stage individually.

The recurring questions were: does the material remember cuts; does its motion agree with its strain; can thin fragments be represented; can contact cover the full volume; and can the complete loop run fast enough for learning?

12.2 Persistent history solved one problem

The first persistent-history prototype kept cut-side information after the blade withdrew. Restricted geometric fixtures showed that separated pieces could move independently and partial cuts retained bridges. This addressed the unwanted reconnection seen in the public CRESSim tests.

However, those cuts were geometry-driven fixtures. A physical-model fracture law still needed to pay for new surface, and blade and fragment contact still needed validation. Bulk-only timing was already a concern.

12.3 Invisible deformation was a deeper failure

A later loaded test found that reconstructed geometry could fold while the stored strain scarcely changed. The force law was effectively blind to some motion. A volume guard detected this, but detecting a failure does not repair the formulation.

An APIC-style repair removed the reproduced invisible mode but introduced or exposed timestep-dependent damping. The next consistent-mechanics formulation kept position, strain and forces tied to one displacement field and avoided repeated transfer filtering. Intact mechanics then passed within its tested envelope.

12.4 Cuts require enough mathematical freedom

A displacement basis describes the shapes of motion the model can express. If both sides of a crack still share the same movement coefficients, the crack cannot actually open. If new coefficients have no properly integrated mass or stiffness, artificial free modes can appear.

Cut-aware integration fixed an off-grid sampling defect. Restricted multicut fixtures then produced independent pieces. Later local-continuity work repaired a different defect where a

shallow partial crack had zero opening rank despite having a geometric interface.

12.5 Accuracy was not just a smaller timestep

Fine crack-front modes can move rapidly. A timestep stable for an intact body may be inaccurate after refinement. The sequence investigated timestep error, activation timing and changes to the representation itself.

A localized event scheme passed a restricted comparison, but at such small timesteps that the performance review rejected it as the laptop's online-learning engine. This was an engineering decision about that implementation and accuracy requirement, not proof that all MPM or cutting simulation is impossible.

12.6 Local mass and selective aggregation

A later branch used a more local representation and linear tetrahedral elements. It repaired an angled-cut mechanical defect but still suffered from very small cut regions, which can force extremely short stable timesteps.

Aggregation ties weakly supported numerical degrees of freedom to better-supported ones. Too much aggregation changed the response beyond the accepted comparison. Selective aggregation passed under an explicitly approved material-weighted velocity measure, while the original stricter pointwise failure stayed documented. That approval was material to the result.

12.7 Real software optimizations improved speed

Caching unchanged geometry, compiling convex operations, reusing GPU arrays and graphs, and overlapping independent preparation with stepping all produced measured improvements. These were meaningful implementation results.

By v22, repeated geometric diagnostic episodes exceeded about 100 aggregate transitions per second in the selected batch. But the benchmark still omitted physical blade contact, paid fracture, robot dynamics and policy inference. It had not yet met the complete-task learning requirement.

12.8 Contact brought the cost back

Adding original-CAD contact exposed sampling and nearest-triangle problems. Refined contact and independent comparisons repaired important component defects. A thin blade is especially sensitive to selecting the wrong nearby triangle or using insufficient numeric precision.

The cost remained high. A later shallow work-paid crack advanced successfully in a tiny coupon, but required repeated geometry and mechanics reconstruction. Optimizations improved exact searches and reduced redundant work without delivering the requested complete fast tomato.

12.9 The v27 decision point

Version 27 ended with the objective explicitly not achieved. It preserved substantial geometry, contact and rollback improvements but did not establish a full fast cut. The concrete choice became whether to continue strict unchanged-physics research or authorize an approximate training model. [S12]

This is the major fork in the project's history. The next model was not a claim that all the strict failures had somehow disappeared. It was a different, explicitly approximate scope.

13. November 13: the approximate model became usable

13.1 What changed in v28

The approximate model used 2,803 massive particles and 31,950 initial elastic bonds, with 3.5 mm spacing and about 121.253 g total mass. Blade motion spent irreversible work to break existing connections; a future fracture plane was not supplied in advance. [S13]

It kept partial bridges, tracked disconnected components after completion, and retained particle contact. The blade was a tapered proxy rather than the original CAD. The particle spheres occupied only about 38% of the former continuum volume, making the spaces between them an explicit approximation.

13.2 What actually passed

The restricted pilot completed work-paid cutting and horizontal withdrawal in all eight benchmark bodies. A subsequent perpendicular cut from an already cut state produced four components, and a larger tomato case also completed. Reset, parallel-state isolation and several stopping conditions were tested.

The final benchmark measured about 136.8 complete task transitions per wall second across eight environments. It included task physics, actions, observations, rewards, termination and selective resets. It excluded articulated robot dynamics, rendering and policy inference. No policy was trained in that benchmark.

13.3 What the pass did not mean

The model's imposed-strain response differed substantially from the old continuum reference. Coarse and fine cutting cycles could finish with the same broken bonds while differing by about 46% in peak force. An oblique smaller-tomato case hit the force stop and was not qualified.

The result justified a restricted approximate-task learning trial. It did not justify claiming realistic force prediction, arbitrary-path robustness, anatomical skin/pulp/seed mechanics, or a complete robot-learning throughput figure.

13.4 Why this is useful anyway

A deliberately approximate environment can be useful for early control experiments if its limits are visible and the controller is later tested against stronger references and reality. It lets the team investigate task interfaces and behavior without waiting indefinitely for a research-grade custom continuum solver.

The risk is learning an exploit of the approximation. That is why future evaluation must include held-out cases, force behavior, stopping rules and external validation rather than accepting a high reward alone.

14. Appearance, speed of motion and the integrated robot

14.1 Smooth appearance was a rendering upgrade

Version 29 added a smooth tomato exterior, stem, calyx and cosmetic flesh/seed colors. The display followed material motion and broken local connections, rather than swapping in a preset sliced tomato. [S14]

The smooth surface visually filled particle gaps; it did not fill them with physical material. Cosmetic seeds were not simulated seeds. The raw particle view remained available so appearance would not conceal the underlying collision representation.

14.2 Faster cutting motion was tested separately

Version 30 changed the controller while preserving the approximate material model. Doubling and quadrupling the feed reduced simulated episode duration in selected cases, but could increase load. A smooth motion variant failed when continuous feed coincided with sliding reversals. [S15]

A synchronized variation improved that particular behavior. These were scripted comparisons, not learned optimal cutting. Timestep sensitivity remained significant, and failed partial runs could not be compared to completed cuts as though their shorter time meant success.

14.3 Version 31 completed a native robot cycle

The approximate material was connected to measured native blade motion in the kitchen scene. The material reaction force and moment were returned once to the native tool, with a declared one-millisecond delay. This was more than moving a rendered blade through a saved animation. [S16]

The successful headless cycle broke 1,857 bonds and produced two components, with about 46% of the mass in the smaller piece. It completed horizontal withdrawal and settling. The saved report records a peak summed simulated load of about 1.3505 N and particle penetration of about 0.0081 mm.

14.4 Important simplifications remained

The knife was held through an ideal fixed attachment after the hand closed. The tomato remained supported through its bottom layer. An ideal retracting cradle cleared the route after pickup. The original tool mass/inertia were retained as a disclosed approximation despite the fitted shorter blade.

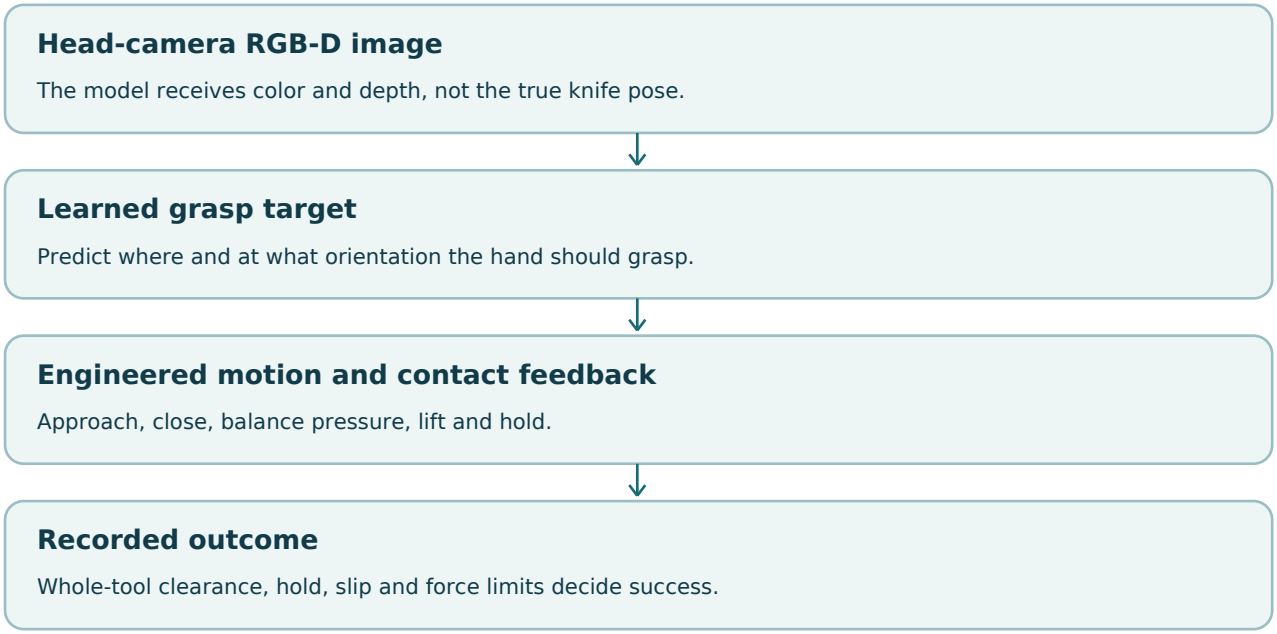
Therefore this was not a demonstrated free-finger grasp carrying cutting loads, nor full dual-arm tomato holding. It was an operational coupled demonstration with clearly described support assumptions.

14.5 The failed integration runs mattered

An early run cut and withdrew but did not settle quickly enough, so its overall result remained failed. A longer bounded settling rule later met the same quiet-state condition. Another run hit a board-contact problem; the contact envelope was corrected while retaining geometric clearance and physical stops.

The successful headless entry/cut/withdrawal/settling section took about 366 wall seconds. A visible GUI cycle also completed but took longer. Neither number was an end-to-end high-throughput robot-learning benchmark. Robot loaded reset and motor-work qualification remained limited.

15. November 13 to 14: learning where to grasp the knife



15.1 The chosen hybrid approach

An initial fully learned image-to-joint approach failed live pickup tests. The user chose a more staged approach: learn the visual grasp position and orientation first, then use a tested motion controller to execute the pickup. Full-motion learning could come later. [S17]

This is a hybrid learned-and-engineered system. The camera model chooses a target; the controller approaches, closes, balances contact and lifts. Calling the entire motion a learned policy would overstate what happened.

15.2 Finger contact replaced the attachment in this separate branch

Unlike the fixed-mount cutting demonstration, these pickup trials used a free knife supported by finger contact and friction. Successful trials had to lift the whole knife clear and hold it unsupported for three seconds.

Closing fingers in the image was not sufficient. Blade clearance, slip, contact load and sustained holding all mattered. This made the result more meaningful than a visually plausible grasp pose.

15.3 The first completed evaluation

The frozen first version passed all three validation placements and seven of eight fresh test placements on the original short-knife setup. One placement lost its grip and remained a failure. A blank-image check rejected the estimate before approach rather than using hidden simulator position. [S17]

That is a useful measured result, but eight tests on one limited setup are not broad statistical evidence for every knife or board position. The two untrained shape probes and opposite-face probe initially failed.

15.4 Why more force was not the answer

Increasing holding effort worsened one slipping case. Recorded contacts showed the fingers twisting the handle against the thumb. Later changes waited for opposing contact, balanced pressure while the board still supported the knife, and used a bounded readiness period before lift.

A grasp must resist a turning moment as well as weight. Pressing harder at the wrong place can rotate the handle out or wedge it against the board. This is why contact location and closing sequence became more important than simply stronger commands.

15.5 Expanding the dataset and cases

The next isolated version covered three knife shapes and both resting faces, with a wider placement range. It collected 240 training images and 12 separate validation images. Fourteen final evaluation cases and an unseen knife variant were reserved. [S19]

Some development grasps were first tested using supplied target positions. Those proved physical feasibility for the controller at those targets, not camera generalization. Five of six shape/face combinations passed such development checks; the heavier knife on its opposite face remained problematic.

15.6 Vision accuracy improved, but grip remained the bottleneck

One expanded model reduced worst validation position error from about 7.9 mm to about 1.4 mm. Later target revisions and a visual selector produced their own measurements; the latest inspected chat reported 1.37 mm maximum error with all 12 saved-image estimates accepted.

These numbers belong to different model/target versions and must not be pooled as one result. Even a one-millimetre placement difference can change which finger surface touches a sloping handle. Accurate image predictions do not guarantee stable contact.

15.7 The latest unresolved experiments

The newer camera-driven comparison produced several successful pickups, but other cases hit force limits, slipped or encountered a blank startup frame. A camera-readiness correction fixed the blank-frame problem without resolving grip retention.

The latest inspected chat reports that finer timesteps and another constraint solver both still encountered force spikes on the heavier flipped knife. It then returned to grasp geometry and began testing load sharing across whichever opposing fingers actually touched, rather than requiring the index finger alone. This was ongoing work, not a completed acceptance result. [S19]

15.8 Why these tests are kept separate from cutting

The pickup prototype has not been shown here to retain the knife under the full cutting load of the integrated tomato demonstration. The integrated cutting demo used a fixed mount. Combining their successes on paper would hide an untested interface between them.

The next bridge is therefore explicit: qualify the free grasp under relevant tool loads, then evaluate the coupled task. That is a future gate, not something a successful unsupported hold already proves.

16. What has actually learned, and what has not

16.1 Resource PPO versus task learning

The October PPO updates were a resource and software exercise with a posture reward. They did not learn pickup or cutting. The November camera model learned visual target prediction. The approximate tomato benchmark supplied an environment suitable for an initial restricted trial but did not train a policy.

These three activities can all contain machine-learning software while having very different meanings. Keeping them separate prevents the phrase 'training worked' from hiding which skill was actually evaluated.

16.2 Observations, actions, rewards and episodes

An observation is what the controller receives, such as an image, tool pose or force signal. An action is its command. A reward is a training score. An episode is one attempt, including a defined reset and termination.

A useful reward cannot compensate for invalid physics. If material reconnects after cutting, a learner can exploit the mistake. If reset leaves damage behind, later episodes start with easier tomatoes. That is why history, masks and resets were developed before claiming learning readiness.

16.3 Privileged information must be disclosed

A simulator knows the exact knife pose and internal damage state. A real camera-equipped robot may not. Such information can be used for training labels or diagnosis, but should not silently enter the evaluated image-only controller.

The newer pickup path explicitly separates teacher-target development checks from camera-driven runs and rejects missing images. This is a methodological improvement even while generalization remains unresolved.

17. Publication, preservation and the paper direction

17.1 The November 14 repository update

The research branch codex/tomato-preparation-20261031 received the newer code, reports, successful and failed evidence, and dashboard snapshots. The main update changed 29,344 files. A follow-up clarified which small vendored source snapshots remained. The verified remote tip was 5f7267f358e8c689ccaa300bd6824ab19436aa2c. [S18]

Large assets, arrays and model weights remained local and were indexed. The original source workspaces and ongoing experiments were not changed by publication. The branch was not merged into main in this work.

17.2 A snapshot is not a portable finished product

Some files are live progress records copied while work is ongoing. Each included file can be hash-verified without the entire collection representing one atomic experiment checkpoint. Machine-specific paths and omitted dependencies still need attention before reproduction.

The repository is valuable as a research archive. It should not be presented as a one-command, calibrated, fully trained system merely because it contains many files.

17.3 How the possible paper contributions narrowed

The discussion considered persistent fracture, arbitrary paths, fast simulation, force-aware cutting, dual-arm coordination, robustness, realistic sensing, sim-to-real transfer and an evaluation framework. These were possible directions, not nine established contributions.

The provisional recommendation was one main contribution: force-aware adaptive cutting. Two supporting directions were robustness across tomato properties and control using realistic observations. The newer pickup work strengthens the perception/contact-control groundwork, but does not itself prove adaptive cutting superiority.

17.4 The research claim still needs experiments

A defensible paper would compare a learned or adaptive controller against a fixed-motion baseline and a well-tuned conventional feedback baseline, then test held-out conditions and sensor ablations. The repository's research rules call for repeated seeds and evidence-backed claims.

If the team instead wants a simulation-method paper, the numerical formulation and its accuracy/performance evidence become the main contribution. That is a different paper scope. The project should not promise both a new high-fidelity simulator and a fully learned dual-arm system without deciding which evidence it can complete.

18. The current boundary and the next logical steps

18.1 What is demonstrated

There are scoped offline material/contact results, native coupling diagnostics, an approximate work-paid cutting task, a fixed-mount robot cutting demonstration, and limited successful camera-guided free-knife pickups. There is also substantial failure evidence explaining why stronger claims are not yet justified.

18.2 What remains unresolved

Real tomato material calibration, high-fidelity complete-cut convergence, general arbitrary-path reliability, fast full-robot training, robust free grasp across knife variants, free-grasp retention during cutting, active dual-arm coordination, motor-work qualification and real-world transfer are not established by this record.

Some components have exact resets; others restart a process or remain only partially tested. Reset claims must therefore name the component rather than treating every branch as one completed environment.

18.3 The practical sequence from here

First, finish the current pickup qualification without mixing teacher-target and camera-driven results or combining different controllers' success counts. Then decide how that grip will be tested under cutting loads. Keep the fixed-mount demo as a separate reference.

For learning, choose a restricted approximate task and declare its limitations. Measure the full loop being trained, compare against conventional control, and reserve held-out cases. For physical realism, obtain appropriate measurements and separate fitting data from validation data.

For writing, begin the system description, decision history and limitations now. Delay strong outcome claims until the matching experiments exist. A clear paper can explain a focused contribution without pretending that every branch of this long project is solved.

18.4 The central lesson of the project so far

The project did not advance along one straight line from a robot scene to a trained cutter. It repeatedly separated things that initially looked equivalent: touching versus holding, damage versus separation, geometry versus mechanics, appearance versus collision, fast kernels versus fast episodes, and simulation success versus real-world validity.

Understanding those distinctions is the most useful way to understand the work. They explain both why progress took many versions and why the later approximate cutting and vision-pickup results should be valued without overstating them.

Appendix A. Glossary

APIC

Affine Particle-In-Cell: a particle/grid transfer approach carrying local velocity variation. The project tested a particular adaptation; its failure does not invalidate the entire method.

Basis / coefficient

A set of building-block motion shapes and their weights. The basis decides what deformations the model can express.

Calibration

Choosing model parameters using measured data. Separate measurements are needed to validate the fitted model.

CAD / collision geometry

The detailed authored shape and the shape actually used to compute contact. They may differ from the rendered display.

Cohesion

Resistance holding opposing material faces together before complete failure.

Connectivity / component

Which material points remain joined. A connected component is a group reachable through remaining material connections.

Consistent mass

An inertia description that retains coupling implied by the displacement representation. A lumped mass is a deliberately simpler local approximation.

Convergence

Checking whether results approach agreement as numerical resolution is refined. Solver convergence, timestep convergence and mesh convergence are different checks.

Coupon

A small specimen designed to isolate a particular physical or numerical behavior.

Damage history

A remembered state describing how much cohesion has been irreversibly lost. With no healing, unloading does not reset damage.

Deformation gradient / J

A local description of how material changes shape. Its determinant J measures local volume ratio; a negative value indicates inversion.

Domain randomization

Varying simulation conditions during training to discourage dependence on one exact configuration. It does not by itself prove realism.

FEM

Finite Element Method: approximate a continuous material using elements and a displacement field, then derive forces and motion.

Fracture energy

Work needed per new cut area in a declared material law. A correct energy counter is not experimental proof of that law.

Gate

A specific acceptance condition that must pass before the next stage. Passing another unrelated check does not override it.

GPU synchronization

Waiting for GPU work to finish before the CPU uses the result. Frequent transfers can make a nominal GPU implementation slow.

Ground truth / teacher target

Exact or deliberately supplied information used for labels or development. It must not silently replace camera inference in an image-only test.

Impedance / compliance

How motion responds to force, often described with spring-like stiffness and damping. Compliance means yielding under load.

Impulse / wrench

Impulse is force accumulated over time. A wrench is force and torque together, expressed at a specified point and frame.

MPM

Material Point Method: material information is carried by points and transferred through a grid. Preventing unintended sharing across cut pieces is a central issue.

Null mode / rank defect

A motion that the discrete model fails to resist or measure as intended. Rank checks help reveal missing mechanical constraints.

Observation mask

A rule limiting which simulator quantities a controller is allowed to see.

PPO / reinforcement learning

PPO is one algorithm for improving a policy using rewards from interaction. Running PPO software does not establish a particular learned skill.

Quadrature

A numerical weighted-sampling rule for integrating force, mass or energy over a region. New thin regions can require new sampling.

Quasistatic

Finding a sequence of approximately force-balanced states while neglecting the full transient dynamics.

Residual policy

A learned correction added to a baseline controller, rather than learning every motion from scratch.

Rollback / transaction

A trial step is accepted as a whole or restored to its previous state, including history and counters.

Simulated time / wall time

Time inside the model versus actual elapsed time on the laptop. Their ratio can be far from one.

Stiffness / damping

Resistance to deformation versus resistance associated with motion. Adding damping can hide instability while changing behavior.

Transition throughput

How many complete environment action steps execute per actual second. Material-only equivalents are not full robot transitions.

Validation / independent test

Validation guides model selection; an untouched test evaluates the frozen choice. Once used for tuning, a test is no longer independent.

Work / energy ledger

A record of input work, stored energy, motion energy and dissipation used to expose missing or double-counted effects.

Appendix B. Version-by-version guide

The v01-v33 identifiers below refer to the isolated November experiment folders. Earlier DiSECt work also used version numbers, so a bare version number without its folder can be ambiguous. This guide preserves that distinction.

v01 - Persistent history

The candidate remembered geometric cuts after withdrawal and passed restricted fragment-independence controls. It had not yet validated physical fracture, contact or useful complete-loop speed.

Evidence: tomato_persistent_mpm_20261110_v01/REPORT.txt

v02 - Physical qualification failure

Reconstructed material could invert while stored strain stayed near its initial value. Finer surface sampling and a whole-volume guard detected the problem; the guard did not cure it.

Evidence: tomato_persistent_mpm_gate3_20261110_v02/REPORT.txt

v03 - APIC transfer repair

The specific invisible-motion control was repaired. Repeated transfer damping and timestep inconsistency still prevented physical qualification.

Evidence: tomato_apic_repair_20261110_v03/REPORT.txt

v04 - Consistent intact mechanics

One displacement field described geometry and strain, with full consistent inertia. Restricted intact tests passed at a very small timestep. Cut geometry and blade contact still needed requalification.

Evidence: tomato_consistent_mechanics_20261110_v04/REPORT.txt

v05 - Restricted cut-aware basis

A partial cut could open while preserving its bridge, and a full division allowed independent pieces. Moving a cut off the sampling grid exposed unsupported numerical modes.

Evidence: tomato_cut_basis_20261110_v05/REPORT.txt

v06 - Cut-aware integration

Integration over cut-created subcells repaired the tested off-grid defect without adding mass or changing material stiffness. It remained a restricted prescribed-topology block test.

Evidence: tomato_cut_integration_20261110_v06/REPORT.txt

v07 - Multiple cuts

Supplied parallel and cross-cut fixtures produced additional independent fragments and retained history. No blade work caused these topology events, and manufactured gaps were used in independence checks.

Evidence: tomato_multicut_20261110_v07/REPORT.txt

v08 - First GPU comparison

A straightforward GPU implementation improved reference speed but left force assembly expensive. The later v09 report preserves the comparison; this is not a complete cutting-task rate.

Evidence: see the following version report and experiment source directory.

v09 - Parallel force evaluation

Parallelizing the same numerical sum produced large relative speedups while retaining reference agreement in tested cases. Absolute runtime for a tiny block remained much too slow.

Evidence: tomato_gpu_comparison_20261111_v09/REPORT.txt

v10 - Current blade sweep creates topology

Successive finite cutting-edge poses generated restricted geometric cuts without a future supplied route. Mechanical evolution ran on GPU, but topology rebuilding still used the CPU.

Evidence: tomato_blade_sweep_20261111_v10/REPORT.txt

v11 - Fine crack front and deformed material

Finer front representation exposed fast local modes and a failed velocity-refinement check. Exact planar mapping also rejected a freely evolving non-affine specimen.

Evidence: tomato_deformed_front_20261111_v11/REPORT.txt

v12 - Finite-face deformed-front correction

The finite CAD cutting face permitted a more conservative geometric mapping. Some finer mechanics checks passed, but combined blade-update refinement still failed.

Evidence: tomato_deformed_face_20261111_v12/REPORT.txt

v13 - Separate timing from spatial changes

Controlled replays showed both event timing and changed motion space contributed to disagreement. A diagnostic mesh using future saved data was not advertised as an online solution.

Evidence: tomato_activation_audit_20261111_v13/REPORT.txt

v14 - Local band and event timing

A preselected local interaction band and precisely localized activation passed a restricted comparison. The blade could not freely leave that band or change plane.

Evidence: tomato_event_front_20261111_v14/REPORT.txt

v15 - Performance no-go

Profiling rejected that high-accuracy consistent-mass formulation as the laptop online-learning backend. This was a cost decision, not a universal impossibility result.

Evidence: tomato_performance_review_20261111_v15/REPORT.txt

v16 - Local-state approximation

Local inertia and trilinear fields reduced coupling complexity, but angled cuts produced extra zero-strain modes. Even intact material-only timing missed the entry target.

Evidence: tomato_local_state_20261111_v16/REPORT.txt

v17 - Linear tetrahedral repair

Linear elements repaired the tested angled-cut rank defect. Very small cut regions constrained the timestep, and the component remained too slow for the intended full task.

Evidence: tomato_linear_local_20261112_v17/REPORT.txt

v18 - Broad aggregation

Constraining weak cut-region degrees of freedom improved a stability diagnostic but changed dynamic response too much. Good initial projection did not guarantee accurate later motion.

Evidence: tomato_aggregation_20261112_v18/REPORT.txt

v19 - Selective aggregation

Only the weakest-supported degrees of freedom were constrained. Component agreement passed under an explicitly approved mass-weighted velocity criterion; larger local errors stayed visible.

Evidence: tomato_selective_aggregation_20261112_v19/REPORT.txt

v20 - Incremental geometry and caching

Unchanged geometry and repeated operators were reused, and GPU assembly improved. The changing-geometry loop became faster without becoming a complete learning environment.

Evidence: tomato_incremental_gpu_20261112_v20/REPORT.txt

v21 - Compiled events and persistent GPU buffers

Compiled clipping and reused GPU graphs reduced event overhead. A complete geometric diagnostic still remained below the preliminary throughput target.

Evidence: tomato_compiled_events_20261112_v21/REPORT.txt

v22 - Overlapped changing-geometry episodes

Independent preparation and stepping overlapped. Repeated synthetic geometric episodes exceeded 100 aggregate transitions/s, before physical contact, fracture, robot and policy costs.

Evidence: tomato_cut_update_redesign_20261112_v22/REPORT.txt

v23 - Original-CAD contact adapter

The material and blade were coupled with rollback and conservative volume checks. Contact sampling had not converged, and dense contact was expensive.

Evidence: tomato_contact_integration_20261112_v23/REPORT.txt

v24 - Contact precision and integration

Wrong nearest-face candidates on the thin blade were corrected with original double-precision triangles. Refined contact tests passed for an intact small cube, not a full tomato cut.

Evidence: tomato_adaptive_contact_20261112_v24/REPORT.txt

v25 - Partial crack cannot open

Tomato geometry and contact were integrated, but a work-paid shallow interface still had zero opening rank. A full-plane control could open, exposing an overly constrained partial representation.

Evidence: tomato_cut_surface_20261113_v25/REPORT.txt

v26 - Local continuity repaired

Partial crack lips acquired independent motion. A shallow CAD-driven paid crack advanced with scoped refinement agreement, but only over a tiny interval and at high cost.

Evidence: tomato_local_continuity_20261113_v26/REPORT.txt

v27 - Exact-search optimizations and scope decision

Hierarchical contact/clearance searches and reduced redundant work improved components. The full fast-tomato objective remained unmet, leading to an explicit approximation decision.

Evidence: tomato_bounded_basis_20261113_v27/REPORT.txt

v28 - Approximate training task

A particle-bond model completed restricted work-paid cuts and batched task loops at about 136.8 transitions/s. Its material error, voids, proxy blade and force sensitivity were disclosed.

Evidence: tomato_approx_rl_20261113_v28/RESULT.txt

v29 - Smooth visual surface

The displayed exterior and exposed interior followed material connections. This added visual clarity, not physical tissue layers or continuum collision volume.

Evidence: tomato_surface_view_20261113_v29/README.txt

v30 - Scripted motion comparisons

Faster feeds shortened selected simulated cuts but increased loads in some cases. Smooth feed at reversal could fail; synchronization helped one tested variant.

Evidence: tomato_motion_comparison_20261113_v30/REPORT.txt

v31 - Native robot integration

Measured robot blade motion drove the approximate tomato and received reaction feedback. A complete cutting cycle passed with an ideal fixed knife mount and supported tomato.

Evidence: tomato_robot_integration_20261113_v31/RESULT.txt

v32 - First visual contact-only pickup

Learned camera targeting plus engineered contact control passed 7/8 fresh short-knife tests. Other knife variants and the opposite resting face initially failed.

Evidence: vision_knife_pickup_20261113_v32/README.txt

v33 - Broader visual pickup, ongoing

Three shapes and two faces were explored with more image data. Teacher-target feasibility and saved-image accuracy improved, but difficult live grips remained unresolved.

Evidence: vision_knife_pickup_generalization_20261114_v33/PLAN.txt

Appendix C. Evidence and reading map

These are primary project records, not a newly verified literature review. Links below open local source files on the project laptop. Many are also preserved in the GitHub archive. Earlier plans can contain expectations later overturned by tests; a plan is not evidence of completion.

Published snapshot: TianjinUniversity research archive, verified November 14 commit

S01 - Early research question and decision log

research_question.md
/home/omen/.gemini/antigravity/scratch/TianjinUniversity/research_state/research_question.md

decisions.md
/home/omen/.gemini/antigravity/scratch/TianjinUniversity/research_state/decisions.md

S02 - Phase audit, laptop benchmark and native integration

PHASE_STATUS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/PHASE_STATUS.md

PHASE2_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/PHASE2_RESULTS.md

NATIVE_INTEGRATION_AND_PHASE3.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/NATIVE_INTEGRATION_AND_PHASE3.md

S03 - Early grasp feasibility and load-support diagnostics

CENTRAL_GRASP_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/CENTRAL_GRASP_RESULTS.md

IN_PLACE_TILT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/IN_PLACE_TILT_RESULTS.md

PALM_SEATING_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/PALM_SEATING_RESULTS.md

CONTACT_SEATING_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/CONTACT_SEATING_RESULTS.md

CONTACT_TRANSFER_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/CONTACT_TRANSFER_RESULTS.md

LOAD_SUPPORT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/LOAD_SUPPORT_RESULTS.md

S04 - Offline material, fracture and contact history

TOMATO_4MM_MESH_COMPARISON_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_4MM_MESH_COMPARISON_RESULTS.md

TOMATO_4MM_REMAINDER_REFINEMENT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_4MM_REMAINDER_REFINEMENT_RESULTS.md

TOMATO_ALL_PATCH_ENERGY_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_ALL_PATCH_ENERGY_RESULTS.md

TOMATO_BLADE_DAMAGE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_BLADE_DAMAGE_RESULTS.md

TOMATO_CLOSED_SURFACE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CLOSED_SURFACE_RESULTS.md

TOMATO_CONTACT_CONTINUATION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CONTACT_CONTINUATION_RESULTS.md

TOMATO_CONTACT_COUPON_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CONTACT_COUPON_RESULTS.md

TOMATO_COUPLED_CONTACT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_COUPLED_CONTACT_RESULTS.md

TOMATO_CUBIC_PATH_DIAGNOSTIC_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CUBIC_PATH_DIAGNOSTIC_RESULTS.md

TOMATO_CURVATURE_SEARCH_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CURVATURE_SEARCH_RESULTS.md

TOMATO_FIRST_FAILURE_STEP_COMPARISON_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_FIRST_FAILURE_STEP_COMPARISON_RESULTS.md

TOMATO_FULL_CUT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_FULL_CUT_RESULTS.md

TOMATO_GLOBAL_SEARCH_DIAGNOSTIC_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_GLOBAL_SEARCH_DIAGNOSTIC_RESULTS.md

TOMATO_INTERFACE_COMPRESSION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_INTERFACE_COMPRESSION_RESULTS.md

TOMATO_LARGE_DEFORMATION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_LARGE_DEFORMATION_RESULTS.md

TOMATO_LAYOUT_AND_BOND_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_LAYOUT_AND_BOND_RESULTS.md

TOMATO_PATCH_ENERGY_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_PATCH_ENERGY_RESULTS.md

TOMATO_SEARCH_RELEASE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SEARCH_RELEASE_RESULTS.md

TOMATO_SLICE_MESH_REFINEMENT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SLICE_MESH_REFINEMENT_RESULTS.md

TOMATO_SPECTRAL_RECLOSURE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SPECTRAL_RECLOSURE_RESULTS.md

TOMATO_SPECTRAL_WITHDRAWAL_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SPECTRAL_WITHDRAWAL_RESULTS.md

TOMATO_SURFACE_MECHANICS_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SURFACE_MECHANICS_RESULTS.md

TOMATO_SWEPT_PATH_DIAGNOSTIC_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SWEPT_PATH_DIAGNOSTIC_RESULTS.md

TOMATO_SWEPT_REFINEMENT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SWEPT_REFINEMENT_RESULTS.md

TOMATO_TRIANGLE_CONTACT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_TRIANGLE_CONTACT_RESULTS.md

TOMATO_TRIANGLE_WITHDRAWAL_RECLOSURE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_TRIANGLE_WITHDRAWAL_RECLOSURE_RESULTS.md

TOMATO_UNLOADING_RECLOSURE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_UNLOADING_RECLOSURE_RESULTS.md

TOMATO_VOLUME_PROTECTION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_VOLUME_PROTECTION_RESULTS.md

TOMATO_VOLUME_SENSITIVITY_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_VOLUME_SENSITIVITY_RESULTS.md

TOMATO_CAPABILITY_AUDIT.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CAPABILITY_AUDIT.md

S05 - Bounded preparation and preserved checkpoint

SOL_PREUPGRADE_15H_PLAN_20261028.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/SOL_PREUPGRADE_15H_PLAN_20261028.md

session.json
/home/omen/teamwork_projects/tomato_cutting_right_arm/runs/offline/tomato_preupgrade_20261027T171031Z/session.json

PROGRESS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/runs/offline/tomato_preupgrade_20261027T171031Z/PROGRESS.md

S06 - Native model integration

physics-materials-integration-20261101.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/physics-materials-integration-20261101.txt

S07 - Research milestone review

RESEARCH_ROADMAP_REVIEW_20261102.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/RESEARCH_ROADMAP_REVIEW_20261102.md

tomato-experiment-data-needed-20261101.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/tomato-experiment-data-needed-20261101.txt

S08 - Research slicing extension

research-slicing-extension-20261106.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/research-slicing-extension-20261106.txt

research-slicing-continuation-20261106.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/research-slicing-continuation-20261106.txt

S09 - Solver feasibility profiling

solver-comparison-20261108.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/solver-comparison-20261108.txt

S10 - DiSECt-derived completion and later status

disect-completion-20261108.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/disect-completion-20261108.txt

disect-section4-current.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/disect-section4-current.txt

DISECT_SECTION5_EXECUTION_PLAN_20261110.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/DISECT_SECTION5_EXECUTION_PLAN_20261110.md

S11 - Public CRESSim feasibility

REPORT.txt
/home/omen/Documents/Codex/experiments/cressim_20261110/REPORT.txt

S12 - Strict numerical sequence and scope decision

REPORT.txt
/home/omen/Documents/Codex/experiments/tomato_bounded_basis_20261113_v27/REPORT.txt

NEXT_SCOPE_DECISION.txt
/home/omen/Documents/Codex/experiments/tomato_bounded_basis_20261113_v27/NEXT_SCOPE_DECISION.txt

S13 - Approximate training model

RESULT.txt
/home/omen/Documents/Codex/experiments/tomato_approx_rl_20261113_v28/RESULT.txt

S14 - Smooth appearance

README.txt
/home/omen/Documents/Codex/experiments/tomato_surface_view_20261113_v29/README.txt

S15 - Motion comparison

REPORT.txt
/home/omen/Documents/Codex/experiments/tomato_motion_comparison_20261113_v30/REPORT.txt

S16 - Integrated fixed-mount robot

RESULT.txt
/home/omen/Documents/Codex/experiments/tomato_robot_integration_20261113_v31/RESULT.txt

S17 - Completed visual pickup evaluation

HYBRID_V07_RESULT.txt
/home/omen/Documents/Codex/experiments/vision_knife_pickup_20261113_v32/reports/HYBRID_V07_RESULT.txt

S18 - Repository archival update

PUBLICATION_UPDATE_20261114.json
/home/omen/teamwork_projects/tianjin-tomato-sync/research/simulation/tomato_cutting_right_arm/PUBLICATION_UPDATE_20261114.json

REPOSITORY_SNAPSHOT.md
/home/omen/teamwork_projects/tianjin-tomato-sync/research/simulation/tomato_cutting_right_arm/REPOSITORY_SNAPSHOT.md

S19 - Latest ongoing pickup branch

PLAN.txt
/home/omen/Documents/Codex/experiments/vision_knife_pickup_generalization_20261114_v33/PLAN.txt

Conversation evidence and scope

The continuation chat titled “Tomato cutting — repository upload and…” (local chat ID 01a191a6-498e-7a40-be2f-fe0df4a0420c) was inspected for the latest pickup work. Its newest turn was still in progress. The latest inspected message described testing load sharing across opposing fingers after alternative-solver and finer-step trials still produced force spikes. This document does not assume those subsequent tests passed.

The current chat supplies the user choices about publication scope, paper-direction discussion, and verified push outcome. The entire lifetime of every chat was not re-read message by message. The account combines those conversations with the accessible saved result documents. Missing dates or undocumented early actions are not reconstructed as facts.

How to use this appendix

Start with the chapter matching your question. Open its source label here, then the result report. If reproducing a number for the paper, follow that report to the raw run, configuration and source hash; check that the quoted case, timestep, controller and model all match. Do not launch old scripts into an existing evidence folder, because some use fixed output names.